



Analysing the suitability of tools used to assess cost-effectiveness of external beam radiation therapy – A scoping review

Taimia Alkanzi^a, Michelle Leech^{a,b,*} 

^a Applied Radiation Therapy Trinity, Discipline of Radiation Therapy, Trinity College Dublin, Ireland

^b Trinity St. James's Cancer Institute, Dublin, Ireland

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ABSTRACT

Background: Cost-effectiveness analyses (CEAs) guide decision-making in radiotherapy. Despite CEAs impact on policy and resource allocation, limited evidence exists on the current tools and methods for external-beam radiotherapy (EBRT). This review aims to identify the suitability of CEA tools utilised in modern EBRT economic evaluations.

Methods: This scoping review followed Joanna Briggs Institute guidelines, with reporting aligned to the Preferred Reporting Items Systematic Reviews and Meta-analyses for Scoping Reviews. A systematic search of PubMed, Embase, and Scopus identified CEA studies on EBRT in adult populations. Data analysis focused on key CEA metrics, including inputs (transition probabilities, time horizons, cycle lengths, discounts, cost, patient factors), outputs (quality-adjusted life years, life-years gained, willingness-to-pay), and sensitivity analyses (deterministic and probabilistic).

Results: Of the 35 identified studies, 86% were model-based CEAs, with Markov being the most common approach. Model-based CEAs often failed to capture disease progression or adverse events adequately. Non-modelling studies relied on incremental cost-effectiveness ratios (ICER) calculations utilising retrospective clinical trial data. No standardised method for addressing missing data was identified. Palliative radiotherapy remains poorly represented, with only five CEA studies addressing it.

Policy summary: This review highlights significant methodological inconsistencies in radiotherapy CEAs, underscoring the need for a standardised approach tailored to EBRT. The breadth of methods and adaptations in conducting CEAs suggests that existing practices do not fully address radiotherapy needs. Despite emphasis on the importance of CEAs in radiotherapy, the evidence base offers little consensus.

1. Introduction

Radiotherapy is an underfunded area of oncology despite established clinical efficacy [1,2]. Inefficient financial structures have created budget and resource allocation gaps, leading to underutilisation [1,3,4]. Underfunding also delays the adoption of advanced radiotherapy techniques [4,5].

Inadequate investments in radiotherapy have caused disparities in access in low- and middle-income countries (LMIC) [6,7]. Conversely, utilisation rates in high-income countries with established radiotherapy infrastructure often fall below recommendations [8–10]. Across Europe, 47%–53.2% of cancer patients are indicated for at least one course of radiotherapy, yet actual utilisation is significantly lower [10]. One study reported that only 40% of non-small cell lung cancer patients receive

radiotherapy compared to the recommended rate of 80% [9].

Radiotherapy implementation delays and underutilisation are compounded by rising cancer incidence and economic pressures. Cancer incidence is expected to rise to 22 million new cases annually by 2030, placing unprecedented demands on healthcare systems worldwide [11]. Simultaneously, annual cancer-related expenditures outpace economic growth and healthcare budgets [12,13], widening the gap between resource availability and patient needs. Vulnerable populations, including older persons and individuals with disabilities, already face barriers to accessing radiotherapy [14,15], which may be further exacerbated as the economic climate worsens. Addressing economic barriers is essential to improving radiotherapy delivery and mitigating health disparities.

Economic evaluations (EEs) guide resource allocation and

* Corresponding author at: Applied Radiation Therapy Trinity, Discipline of Radiation Therapy, Trinity College Dublin, Ireland.

E-mail address: leechm@tcd.ie (M. Leech).

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prioritisation in radiotherapy [16–18]. Providing an objective cost-assessment tool is important as radiotherapy is often perceived as expensive despite evidence supporting its value [10,19,20]. Value-based cancer care is becoming increasingly relevant to decision-makers and patients engaging with private or semi-private healthcare systems [21, 22]. EEs assessing costs against benefits can mitigate some barriers to radiotherapy by assessing the financial impact of radiotherapy and improving the perceptions of physicians and stakeholders. Yet, current EE methodologies lack standardisation compared to pharmacoeconomics [16,23,24].

Several EE methods are used in oncology, including micro-costing, cost-comparisons, and cost-effectiveness analyses (CEAs) [16,25]. While micro-costing and cost-comparisons are practical costing techniques, they are less applicable to radiotherapy, as both methods exclude patient-related factors, namely overall survival (OS), progression-free survival (PFS), or adverse effects [26,27].

CEAs evaluate interventions by integrating economic and patient-related data. Health effects are quantified as quality-adjusted life years (QALYs) or life-years gained (LYG) by incorporating OS, PFS, adverse effects, or health-related quality-of-life (HR-QoL). Ideally, radiotherapy CEAs would include all quantifiable health data (e.g. loss of productivity, fatigue); however, data are often incomplete or low-quality [28].

Economic data should include the costs of equipment acquisition, infrastructure, and staff training to avoid underestimating financial needs [29]. Cost underestimation can result in treatment implementation delays and insufficient funding [30]. In LMICs, undervaluing costs can lead to poor investment prioritisation, further restricting access to radiotherapy. Meanwhile, in private and semi-private healthcare systems, it can raise patients' out-of-pocket expenses, leading to radiotherapy underutilisation [31].

Outcomes are reported as incremental cost-effectiveness ratios (ICERs) and assessed against a willingness-to-pay (WTP) threshold [32]. Historically, the WTP threshold was set at 50,000/QALY(€/£), yet recent CEAs have moved towards a threshold of 100,000/QALY(€/£) due to rising treatment costs [33–35]. Currently, little consensus exists on a radiotherapy-specific threshold.

To date, no review has evaluated the methodological robustness of CEAs in radiotherapy. Previous reviews focused on systematic therapies [36–38] or assessed reporting inconsistencies without analysing the impact of reported metrics [23]. Other reviews in radiotherapy either broadly assessed all EEs, were site-specific, treatment-specific, or country-specific [24, 39–41].

Given CEAs role in appropriate resource allocation and radiotherapy accessibility, evaluating the current landscape of EEs in radiotherapy is essential. The suitability of current CEA methods for external-beam radiotherapy (EBRT) specifically requires clarification. Suitability of CEA methods in the context of EBRT refers to their ability to provide rigorous, transparent and context-relevant data to guide resource allocation, by comparing costs of EBRT to patient outcomes. Suitability must also consider the data inputs and sensitivity analyses of various methods [16].

This scoping review therefore aims to identify the suitability of CEA tools for modern EBRT economic evaluations. The review will focus on inputs (transition probabilities, time horizons, discounts, cost, patient factors) and outputs (QALYs, LYG, WTP). It will also define and analyse the sensitivity analyses (deterministic and probabilistic) used in the identified studies.

2. Methods

This scoping review was conducted according to Joanna Briggs Institute (JBI) guidelines [42] and further refined using JBI guidelines for reviewing economic evaluations [43]. To ensure methodological rigour, the review follows the Preferred Reporting Items Systematic Reviews and Meta-analyses for Scoping Reviews (PRISMA-ScR) guidelines [44].

2.1. Search strategy for identification of studies

The literature search was conducted systematically in three electronic databases (Embase, PubMed, and Scopus). The databases were selected for their comprehensive coverage of healthcare and economic research. Relevant studies were identified using keyword combinations and Boolean operators. Key search terms included 'cost-effectiveness analysis', 'cost-utility analysis', 'economic evaluations' and 'health economics'. (Appendix A details the specific search terms used per database).

The three databases identified 2313 papers, which were exported to the reference manager, Endnote. Duplicate studies were then removed. The remaining papers were screened by title and abstract. Studies that reflected a paediatric population, non-radiotherapy or non-EBRT interventions were removed. At initial screening, 180 studies were assessed for eligibility by full-text review. To ensure consistency in applying the inclusion and exclusion criteria, the full-text screening process was piloted on a subset of studies before the full review. The final number of studies included in this review is 35. No additional studies were identified from screening reference lists. The PRISMA flowchart depicts the identification and screening process (Fig. 1).

2.2. Type of studies

Full-text studies published in English between 2009 and 2024 were included. The publication dates were limited to focus on recent studies relevant to modern EBRT, while ensuring any seminal papers were not excluded. Conference abstracts, systematic reviews, meta-analyses, literature reviews, editorials, and letters to editors were excluded.

2.3. Type of participants

Paediatric and non-human studies were excluded due to differences in interventions, economic data and outcomes. Studies reporting on non-radiotherapy (pharmaceutical, surgical, holistic) and non-EBRT (brachytherapy, intra-operative radiotherapy) interventions were excluded to reduce heterogeneity in study selection.

2.4. Type of interventions

Studies that did not conduct a CEA or include non-cost data were excluded.

2.5. Type of outcomes and data analysis

The outcomes examined were QALYs, LYG, and WTP. Studies that reported one of these metrics were included as they are standard metrics reported in health economic evaluations. The methods and results of deterministic and probabilistic sensitivity analysis were analysed.

2.6. Assessing the quality of studies

The quality of the included studies was assessed using the Drummond checklist [45]. The ten-item checklist guides the evaluation of health economic studies and allows for assessing EEs for risk of bias and methodological rigour. A rating scale, attributing one point per answer, assigned a total score to each study [46].

3. Results

The review identified 35 relevant studies investigating cost-effectiveness across several radiotherapy planning and delivery techniques. Tables 1 and 2 summarise the CEA study parameters and outcomes.

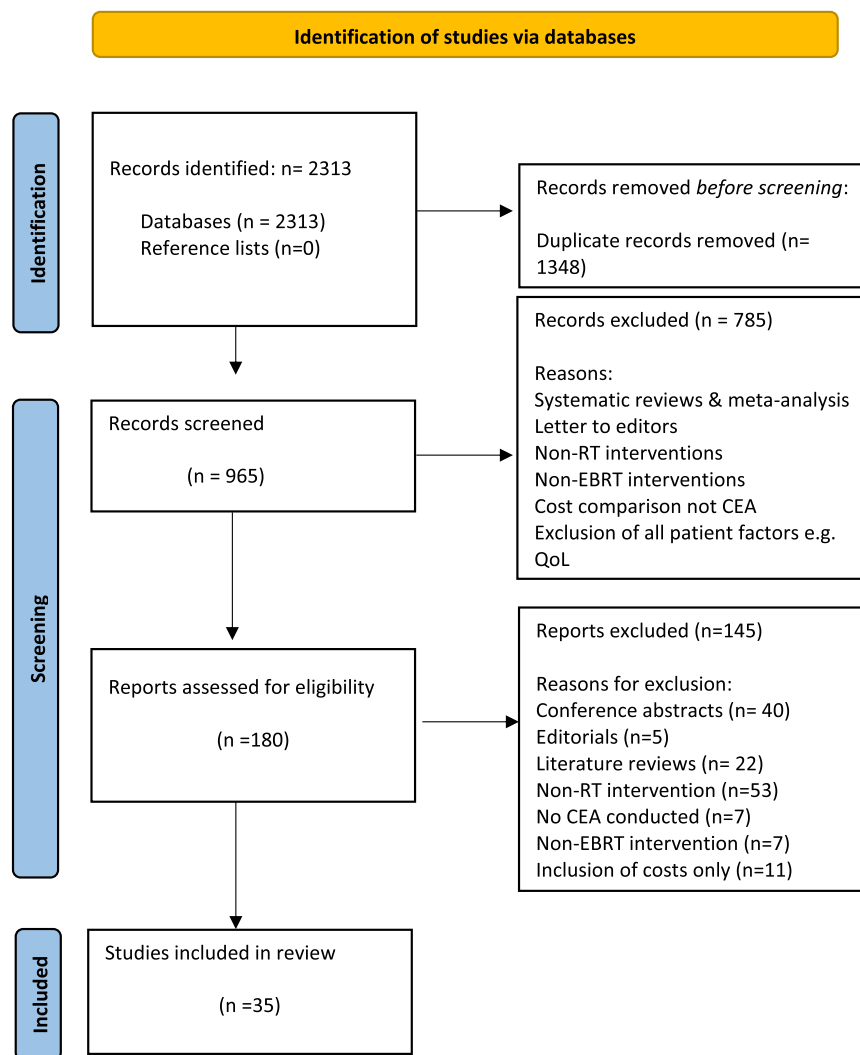


Fig. 1. Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) flowchart. Identification and screening of selected studies. Abbreviations: n = number of, RT= radiotherapy, EBRT= external beam radiotherapy, CEA= cost-effectiveness analysis, QoL= quality of life.

3.1. Quality of included studies

The results of the Drummond checklist and the rating scale provide an assessment of poor (1–3 points), average (4–7 points) or good (8–10 points). All studies scored between 4 and 10 points (i.e. average or good). No studies were excluded due to poor methodological quality (Appendix B). Studies deemed ‘average’ using this quality assessment typically failed to credibly value costs and consequences [49,58,60,70, 71,73,75,76]. Some did not include any uncertainty estimates [49,52, 70,75,78] or adjust costs and consequences for differential timing [52, 58,73,76,78]. One did not make allowance for uncertainty estimates nor discuss all results [47].

3.2. Study parameters

3.2.1. Model type/ methodology

CEAs can be model-based or non-model-based. The method applied determines the study parameters and, therefore, affects the outcomes. This review identified 28 model-based [47–74] and 7 non-model-based CEAs [75–81]. (Fig. 2). Of the 28 model-based studies, 23 (66%) used a Markov model [47–51, 54, 55, 58–63, 65–69, 71–74, 82]. The remaining models included one non-specific decision analytic model [53], one two-part regression model [56], one decision tree model [57], one state transition microsimulation model [70], and one combined decision tree

and Markov model [64]. The seven (20%) non-model-based CEAs used an ICER equation (75–81). A direct calculation was possible due to the availability of retrospective patient data from published randomised clinical trials (RCTs).

The variation of methodologies suggests a need for standardisation within radiotherapy CEAs. Consistent with the literature, this review found that Markov models were the most common method identified, however, Markov models cannot capture all aspects of disease progression across prolonged periods [83,84]. Similarly, non-modelling CEAs do not account for some aspects of the patient experience but rely on trial data.

The decision analytic model applied by one study is a non-specific description of the model [53]. Markov models, decision trees, and microsimulations can be described as decision analytic models. Thus, this description lacks clarity and indicates poor reporting, compromising the study’s replicability.

One study’s microsimulation model facilitated assessing workflow timing simultaneously [70]. A model that accounts for workflow times allows for realistic conclusions on a new intervention’s practical applicability. This becomes important in high-intensity work environments caused by staff shortages and underfunding [85,86]. No other study included workflow times.

Table 1

Study Characteristics. Abbreviations: HFRT= hypo-fractionated radiation therapy, CFRT= conventionally fractionated radiation therapy, QALYs= quality-adjusted life years, CE= cost-effective, OS= overall survival, FIF= field-in-field, IMRT= intensity modulated radiation therapy, STD-VMAT= standard volumetric modulated arc therapy, MA-VMAT= multiple arc volumetric modulated arc therapy, NC-VMAT= non-coplanar volumetric modulated arc therapy, 3DCRT= 3 dimensional conventional radiation therapy, IMRT-S= intensity modulated radiation therapy with spacer, NR= not reported, SBRT= stereotactic body radiation therapy, CIRT= carbon ion radiation therapy, PBT= proton beam therapy, PFS= progression-free survival, WTP= willingness-to-pay, MRgRT= magnetic resonance-guided radiation therapy, CT-IGRT= computed tomography image-guided radiation therapy, HA-WBRT= hippocampal avoidance whole brain radiation therapy, SRS= stereotactic radiosurgery, WBRT= whole brain radiation therapy, PCI= prophylactic cranial irradiation, HA-PCI= hippocampal avoidance prophylactic cranial irradiation, ART= adaptive radiation therapy, LYG= life-years gained, HRS= hydrogel rectal spacer, EF= erectile function, 2DCRT= 2 dimensional radiation therapy, DIBH= deep inspiration breath hold, FB= free-breathing, RT= radiation therapy, IMPT= intensity modulated proton therapy, VMAT= volumetric modulated arc therapy.

Authors [Reference]	Year of publication	Intervention	Control	Reimbursement type	Probabilistic methodology	Number of simulations	Result of the probabilistic analysis	Type of sensitivity analysis	Result of sensitivity analysis
Zhou et al.	2022	HFRT	CFRT	Public	Bootstrap	1000	97.8% HFRT	Generalised linear model	Sensitive to QALYs
Yong et al.	2012	IMRT	3DCRT	Public	X	X	X > 50% HFRT CE (China)	One-way sensitivity analysis	Sensitive to cost of IMRT and frequency of grade 3 gastrointestinal events
Yang et al.	2021	HFRT FIF, hybrid-IMRT, full-IMRT, STD- VMAT, MA- VMAT, NC-VMAT	CFRT	Private/ public	Monte Carlo	10,000	100% HFRT CE (France & US)	One-way sensitivity analysis	Sensitive to OS
Xie et al.	2021		Conventional tangential fields	Private	Monte Carlo	100,000	FIF 58.9% at 50,000/QALY MA-VMAT 72.6% at 100,000/QALY	One-way sensitivity analysis	Sensitive to probability to developing contralateral breast cancer
Vanni et al.	2021	IMRT	3DCRT	Private	X	X	X	One-way sensitivity analysis	Sensitive to cost of intervention to treat severe toxicity
Vanneste et al.	2015	IMRT-S	IMRT only	Public	NR	NR	77% IMRT-S CE	One-way sensitivity analysis Generalised linear model	Sensitive to healthy utility figure
Sun et al.	2023	Ultra-hypo fractionated RT (UHRT)	CFRT	Public	Bootstrap	1000	No significant differences between both arms	Modified Park test Ordinary Least Square method	Sensitive to costs
Sprave et al.	2018	CIRT	PBT	Public	X	X	X	One-way sensitivity analysis	Sensitive to the 10-year PFS in CIRT
Sher et al.	2014	SBRT (robotic-SBRT & nonrobotic-SBRT)	IMRT	Private	Monte Carlo	NR	SBRT likely to be CE at \$100,000/ QALY WTP	One-way sensitivity analysis	Sensitive probability of transitioning from no evidence of disease to local recurrence and to increased toxicity
Shah et al.	2019	Accelerated partial breast irradiation	Hypofractionated whole breast irradiation (with or without boost)	Private	X	X	X	One-way sensitivity analysis	Sensitive to frequency of acute toxicity
Schumacher et al.	2020	MRgRT HA-WBRT SRS + WBRT SRS + HA-WBRT SRS + salvage (SRS /HA-WBRT/ WBRT)	CT-IGRT	Public	X	X	X	One-way sensitivity analysis	Sensitive to number of patients treated
Savitz et al.	2015		WBRT	Public	Monte Carlo	1000	At \$43,000/QALY WBRT-49.3% HA-WBRT: 47.7% At \$108,000/QALY SRS+HA-WBRT: 46.1% HA-WBRT: 45.1%	One-way sensitivity analysis	Sensitive to neurocognitive impairment
Sampaio et al.	2024	PBT	CFRT	Public	Bootstrap	5000	< 30% PBT CE	One-way sensitivity analysis	Most sensitive factor not identified

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Table 1 (continued)

Authors [Reference]	Year of publication	Intervention	Control	Reimbursement type	Probabilistic methodology	Number of simulations	Result of the probabilistic analysis	Type of sensitivity analysis	Result of sensitivity analysis
									They tested QALYs, utilities and data completeness
Richard et al.	2016	IMRT	3DCRT	Public	Monte Carlo	1000	64% IMRT CE 68% HA-PCI CE at \$100,000/QALY WTP threshold	One-way & two- way sensitivity analysis	Sensitive to probability of local recurrence
Qu et al.	2017	Hippocampal avoidance PCI	Conventional PCI	Private	Monte Carlo	5000	48% HA-PCI CE at \$50,000/QALY WTP threshold 75.1% IMRT CE at 50,000/QALY WTP threshold	One-way & two- way sensitivity analysis	Sensitive to relative risk of new brain metastases
Parthan et al.	2012	SBRT PBT Scheduled weekly	IMRT	Private	Monte Carlo	NR	94.1%-95.5% PBT CE at 50,000/ QALY WTP threshold	One-way sensitivity analysis	Results were not sensitive to variables tested
Perrier et al.	2024	ART	Standard IMRT	Public	Bootstrap Monte Carlo with triangular distributions	NR	70% ART CE at high WTP threshold	One-way sensitivity analysis	Sensitive to number of patients with xerostomia
Okazaki et al.	2022	CIRT	SBRT	Private		10,000	At lower WTP 22.1% At higher WTP 50.3% for CIRT	One-way sensitivity analysis One-way & two- way sensitivity analysis	Sensitive to LYG in CIRT cohort
Mitera et al.	2014	SBRT	CFRT	Public	X	X	X	One-way sensitivity analysis	Sensitive to survival rates and costs
Marta et al.	2018	IMRT	3DCRT	Public	Monte Carlo	1000	90% IMRT CE Without coronary risk factors PBT is not CE at lower threshold and only CE in certain mean heart dose at the higher WTP threshold	One-way sensitivity analysis	Sensitive to utility of low xerostomia & dysphagia
Mailhot et al.	2016	PBT	CFRT	Private	Monte Carlo	50,000		One-way sensitivity analysis	Sensitive to mean heart dose
Li et al.	2022	PBT	CFRT	Private	Monte Carlo	50,000	NR 44.21% HRS CE in hospital setting composed of the following: 71% HRS CE for men with good EF at baseline 6.6% HRS CE for men with poor baseline EF	One-way sensitivity analysis	Sensitive to the cost of PBT
Levy et al.	2019	CFRT + HRS	CFRT	Private	Monte Carlo	10,000		One-way sensitivity analysis	Sensitive to utility of erectile dysfunction
Leung et al.	2017	PBT	SBRT	Public	Monte Carlo	10,000	97% PBT CE	One-way sensitivity analysis	Sensitive to utility and costs of both treatments
Lester-Coll et al.	2016	SRS (SRS + WBRT)	WBRT (SRS+WBRT)	Private	Monte Carlo	100,000	38% SRS CE at primary WTP, 67% CE at secondary WTP threshold 90% SRS CE at \$50,000/QALY	One-way sensitivity analysis	Sensitive to SRS costs
Kim et al.	2017	SRS + WBRT	SRS alone	Private	Monte Carlo	NR		One-way sensitivity analysis	Sensitive to probability of cognitive decline in SRS
Kerba et al.	2024	SBRT	CFRT	Public	Monte Carlo	10,000	NR HRS for all patients (AUD): 3% at \$28,000 WTP 10% at \$42,000 WTP 25% at \$67,000 WTP HRS for select patients (AUD): 38.3% at \$28,000 WTP 62.6% at \$67,000 WTP	One-way sensitivity analysis	Sensitive to utility of postprimary treatment
Jones et al.	2021	CFRT+ HRS	CFRT alone	Public	Monte Carlo	10,000		One-way sensitivity analysis	Cost of hydrogel spacer (continued on next page)

Table 1 (continued)

Authors [Reference]	Year of publication	Intervention	Control	Reimbursement type	Probabilistic methodology	Number of simulations	Result of the probabilistic analysis	Type of sensitivity analysis	Result of sensitivity analysis
Glynn et al.	2023	HFRT (partial breast & whole breast)	CFRT (partial breast & whole breast)	Public	Monte Carlo	10,000	62% partial breast HFRT CE compared to CFRT 100% whole breast HFRT CE compared to CFRT 0.9% IMRT CE compared to 2DCRT 8.4% IMRT CE compared to 3DCRT 2.3% 3DCRT CE compared to 2DCRT	One-way sensitivity analysis	Sensitive to applying hazard ratio for distant recurrences
Chauhan et al.	2020	IMRT (3DCRT)	2DCRT (3DCRT)	Public	Monte Carlo	999		One-way sensitivity analysis	Sensitive to IMRT costs Sensitive to cost of linac & excess cardiovascular disease rate ratio
Buschaert et al.	2024	DIBH RT	FB RT	Public	Monte Carlo	10,000	99.85% DIBH CE	One-way sensitivity analysis	Sensitive to cost of linac & excess cardiovascular disease rate ratio
Brodin et al.	2021	IMPT	IMRT	Private	Monte Carlo	10,000	42.4% IMPT CE	One-way sensitivity analysis	Sensitive to cost of IMPT
Austin et al.	2019	IMPT	VMAT	Hybrid (public/private)	X	X	X	One-way sensitivity analysis	Sensitive to PBT cost Sensitive to secondary cancers parameters (costs, utility, death probability)
Austin et al.	2021	IMPT + DIBH	IMRT + DIBH	Hybrid (public/private)	X	X	X	One-way sensitivity analysis	Sensitive to local recurrence treatment in the SBRT arm
Arruda et al.	2022	SBRT	CFRT	Public	Monte Carlo	10,000	92% SBRT CE	One-way sensitivity analysis	SBRT arm

3.3. Study inputs

3.3.1. Transition probabilities

Transition probabilities are essential to model-based CEAs. All model-based CEAs are required to move through the model at set time intervals [84]. The transition probabilities determine the simulated transition process across health states (e.g. metastatic disease to death) and can be sub-categorised as constant or time-varying. Constant transition probabilities use annual/monthly hazard rates and apply those rates consistently, as observed in 9 studies [51,53,54,59,65,67,69,74, 82]. Time-varying transition probabilities vary across several factors, including age, quality-of-life (QoL), and radiotherapy dose, as reflected by 12 studies [55, 58, 60–62, 64, 66, 68, 70–73]. The remaining 11 studies did not report or did not use transition probabilities [47, 50, 56, 57, 75–81]. However, non-model-based CEAs do not apply transition probabilities as the availability of trial outcomes allows them to bypass simulating patient data.

3.3.2. Time horizon & cycle length

The time horizon is a defined duration over which cost and health outcomes are assessed [87]. Several factors can influence the time horizon of a CEA, including the intervention, treatment intent and patient cohort. The selected studies displayed a variety of time horizons; the 5-year timeframe was the most common (27%, N = 9) [50,51,55,57,62, 63,65,67,69]. Other studies used a lifetime horizon (26%, N = 9) [47, 48,60,61,66,71,72,74,82], 15 years (9%, N = 3) [49,54,59] and 2 years (6%, N = 2) [64,78]. Of the remaining 12 studies, 11 used varied time horizons [53, 56, 58, 68, 70, 73, 75–77, 79, 80], and one did not report the time horizon [81].

Patients may experience late effects (e.g. cardiotoxicity, secondary cancers) up to 20 years post-treatment [88,89]. Therefore, incorporating some late effects is challenging when using shorter time horizons. The shortest time horizon identified was 90 days; the authors argued that no statistically significant difference existed between the intervention and control beyond 90 days [53]. There was no consideration for economic differences across a longer period.

The cycle length is closely related to the time horizon. It defines fixed periods within a model during which a patient can transition between health states (e.g. local recurrence to metastatic disease) or remain in the same health state. Non-model-based CEAs do not require cycle lengths due to the availability of retrospective patient data. Most model-based CEAs applied annual cycles (N = 10) [49, 51, 54, 60, 61, 68, 70–73]. The remaining studies reported cycles ranging from six months to monthly [48, 50, 55, 58, 62–67, 82]. One study applied three-month cycles for the first year and annual cycles thereafter [69]. The cycle length aligns with standard follow-up timelines, possibly reflecting staff-associated costs more accurately. Five model-based CEAs did not report cycle lengths [53,56,57,59,74].

Long cycles (e.g. 1 year) in short time horizons (e.g. 5 years) cannot accommodate disease complexity (e.g. progression, adverse events) [90]. Patients with cancer can experience multiple events simultaneously. Model-based CEAs inherently simplify this process by assuming one transition per cycle. Vanneste et al. [51] adopted an annual cycle for a 5-year time horizon. The study includes acute and late gastrointestinal effects; however, it would be challenging to assess acute toxicities accurately if only one transition can occur per cycle. Acute toxicities can progress through grades over one to two weeks [91]. Thus, the annual cycle length for this study should be revised.

Conversely, shorter cycle lengths improve accuracy but cause a high computational burden [92]. Two studies paired a lifetime horizon with monthly cycles [47,48]. The former study was based on the trial NCT00793962 [93], which conducted frequent follow-ups and produced sufficient clinical data, facilitating monthly cycles and reducing the computational burden of a shorter cycle length.

Table 2

Study Inputs & Outputs. Abbreviations: CEA= cost effectiveness analysis, EE= economic evaluation, HR-QoL= health related quality of life, ICER= incremental cost effectiveness ratio, QALY= quality adjusted life years, LYG= life years gained, WTP= willingness to pay, GI= gastrointestinal, GU= genitourinary, CE= cost-effective, OS= overall survival, RCT= randomised control trial, 3DCRT= 3 dimensional conventional radiation therapy, IMRT-S= intensity modulated radiation therapy with spacer, NR= not reported, SBRT= stereotactic body radiation therapy, CIRT= carbon ion radiation therapy, PBT= proton beam therapy, DFS= disease free survival PFS= progression-free survival, MRgRT= magnetic resonance-guided radiation therapy, CT-IGRT= computed tomography image-guided radiation therapy, HA-WBRT= hippocampal avoidance whole brain radiation therapy, SRS= stereotactic radiosurgery, WBRT= whole brain radiation therapy, PCI= prophylactic cranial irradiation, HA-PCI= hippocampal avoidance prophylactic cranial irradiation, ART= adaptive radiation therapy, HRS= hydrogel rectal spacer, EF= erectile function, 2DCRT= 2 dimensional radiation therapy, DIBH= deep inspiration breath hold, FB= free-breathing, RT= radiation therapy, IMPT= intensity modulated proton therapy, VMAT= volumetric modulated arc therapy.

Authors [Reference]	CEA model/ Method of EE	Time Horizon/ Cycle Length	Input: Cost data	Input: Discoun- ting	Input: Transition Probabilities	Input: Adverse Effects (acute/ late)	Input: HR- QoL	Input: survival	Output: ICER	Output: QALY	Output: LYG	Output: WTP
Zhou et al.	ICER Equation	4 years/ X	Direct indirect medical cost, direct nonmedical cost	2.5% costs & HR-QoL	X	Acute & late GI & GU effects Fisher's exact test	EPIC-50 EQ-5D-3 L	Individual patient data from French arm of RCT	NR	CFRT 3.252 Vs HFRT 3.297	X	€50,000/ QALY
Yong et al.	Markov model	Lifetime/ 1 month	Direct medical costs and indirect medical costs	5% costs & QALYs	NR	Acute & late grade 2 & 3 GI effects	X	RCT OS assumed the same in both arms	\$26,768/ QALY	IMRT 6.085 3DCRT 6.062 China: CFRT 13.1254 HFRT 13.1091 France: CFRT 13.4244 HFRT 13.4125 US: CFRT 13.8036 HFRT 13.8009	X	NR
Yang et al.	Markov model	Lifetime/ 1 month	Direct medical costs Direct & indirect medical costs	3% costs & QALYs	Time dependant hazard ratios LAR & RCE for annual hazard ratios	Acute & late effects incorporated with general utility value Late effects: Secondary cancers & cardiotoxicity LAR & RCE	EQ-5D	RCT patient data Weibull survival function (HFRT) log-normal survival function (CFRT)- both fitted onto a Kaplan-Meier graph Web-based calculator of breast outcomes (cancer-math)	China: \$144,281.47/ QALY France: \$420,636.10/ QALY US: \$3187,955.76/ QALY	Stratified by treatment technique	X	\$30,828/ QALY (China) \$100,000/ QALY (US, France) €50,000/ QALY €100,000/ QALY 100,000/ QALY 50,00/ QALY Brazilian
Xie et al.	Markov model	15 years/ 1 year	Direct & indirect medical costs	3% costs& utilities			X				X	
Vanni et al.	Markov model	5 years/ 1 month	Direct & indirect medical costs	5% Costs & QALYs	NR Constant annual transition probabilities	Acute & late grade 2 + GI & GU effects	X	RCT OS assumed the same in both arms	R\$ -296.74/QALY	IMRT= 41.08 3DCRT= 19.25	X	
Vanneste et al.	Markov model	5 years/ 1 year	Direct & indirect medical costs (HCIB) Direct & indirect medical costs, direct non-medical costs	4% costs & 1.5% QoL		Acute & late GI effects	EQ-5D	OS assumed the same in both arms	€55,880/ QALY SEK/QALY (-3913,4 87,2309,086)- cost1 (-15,818,470, 11,743,449)- cost2	IMRT-S= 3.570 IMRT-O= 3.542	X	€80,000/ QALY
Sun et al.	ICER equation	6 years/ X	Direct medical costs (single institution cost analysis)	3% costs & QALYs	X	Acute & late GI effects, fatigue, pain & sleep disturbance	EORTC-QLQC30 mapped to EQ5D-3L	Individual patient data from an RCT PRT OS pooled 5- & 10-year gamma knife data CIRT OS data NR		-0.1315 CI-low 0.0273 Clup	X	NR €105,135/Q ALY- high CE €35,045/QA LY-CE (WHO)
Sprave et al.	ICER equation	34 years/ X			X	X	X		€8855.76/ QALY	8.26 (CIRT) 6.65 (PRT)	X	

(continued on next page)

Table 2 (continued)

Authors [Reference]	CEA model/ Method of EE	Time Horizon/ Cycle Length	Input: Cost data	Input: Discoun ting	Input: Transition Probabilities	Input: Adverse Effects (acute/ late)	Input: HR- QoL	Input: survival	Output: ICER	Output: QALY	Output: LYG	Output: WTP
Sher et al.	Markov model	Lifetime/ 4 months	Direct medical costs (2012 Medicare publication, literature)	3% costs & QALYs	Constant hazard ratios	Acute & late GU, rectal & sexual effects within 5 years (urinary, sexual) Rectal	X	OS data from consensus and review literature	R-SBRT: \$285,000/QALY NR-SBRT: \$591,100/QALY	R-SBRT: 9.93 NR-SBRT: 9.93 IMRT: 9.96	X	\$100,000/QALY
Shah et al.	Decision analytic model	90 days/ NR	Direct, indirect medical costs and direct nonmedical costs (Medicare)	2.5% cost of wages	Constant transition probabilities	Acute grade 3 + late toxicity assumed the same in both arms	X	OS assumed the same in both arms	Not reported detailed	Whole breast HRT (both): 0.2289 Accelerated HRT: 0.2300	X	NR \$50,000/QALY \$100,000/QALY
Schumacher et al.	Markov model	15 years/ 1 year	Direct, indirect medical costs (Medicare, literature)	3% costs & utilities	Constant hazard ratios	Acute & late GI & GU grade 2 + effects	X	OS assumed the same in both arms	X	NR	X	\$100,000/QALY
Savitz et al.	Markov model	5 years/ 1 month	Direct & indirect medical costs (Medicare)	3% costs & effects	Time dependant transition probabilities	Acute & late Neurocognitive decline & fatigue	X	Model 4 median survival lengths (3, 6, 12, 24 months)	Stratified by month and by intervention(s)	Stratified by months and by intervention(s)	X	\$100,000/QALY
Sampaio et al.	Two-part regression model (probit & GLMs)	58 weeks/ NR	Direct & indirect medical costs (Swedish national registries)	X	NR	X Acute & late grade 2 + effects:	EORTC-QLQC30 with QLUC10D algorithm	Based on observational study	ICER for each patient calculated separately on the CE plane	PBT: 0.746 CRT: 0.774	X	NR
Richard et al.	Decision Tree model (rollback analysis)	5 years/ NR	Direct & indirect medical costs (literature & Medicare)	X	NR	oedema, joint pain, fracture, adverse wound events and fibrosis	X	OS based on 3 clinical studies and calculated as utilities	ICER for IMRT: \$16,842/QALY	IMRT: 3.68 3DCRT: 3.35	X	\$50,000/QALY \$50,00/QALY \$100,000/QALY (primary)
Qu et al.	Markov model	8 years/ 3 months	Direct medical costs (literature, Medicare)	3% costs & QALYs	Varying transition probabilities	Late effect: Neurocognitive dysfunction	X	Digitised Kaplan-Meier graph from published studies	HA-PCI ICER: \$47,107/QALY	HA-PCI: 1.85 C-PCI: 1.75	X	\$50,00/QALY \$100,000/QALY
Parthan et al.	Markov model	Lifetime/ NR	Direct Non-medical costs	3% costs & utilities	Constant transition probabilities	Late grade 2 + GI GU & EF effects	RCT published data	OS assumed the same in both arms	NR €194,521 per xerostomia avoided at 24 months €162,444 per xerostomia	IMRT: 8.05 SBRT: 8.11 PBT: 8.06	X	\$50,00/QALY \$100,000/QALY
Perrier et al.	ICER equation	2 years/ X	Direct medical costs (French national health data system)	2.5% costs	X	Xerostomia measured at 12 & 24 months	X	OS & PFS based on RCT data	7491,017 JPY/LY 3708,330 JPY/LY	X	X CIRT:4.5 15 SBRT: 3.358 SBRT: 1.03 years	NR 4000,000/ LYG (JPY) 8000,000/ LYG (JPY) \$50,000/ QALY
Okazaki et al.	ICER equation	4 weeks & 5 years/ X	Direct medical costs (retrospective data)	X	X	X	X	OS: Kaplan Meir estimates	\$1156/LYG for SBRT	X		
Mitera et al.	ICER equation	Survival period up to death or	Direct medical costs (retrospective	X	X	Acute & late: Acute grade 3 + events:	X	OS: Kaplan Meir estimates				

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Table 2 (continued)

Authors [Reference]	CEA model/ Method of EE	Time Horizon/ Cycle Length	Input: Cost data	Input: Discoun ting	Input: Transition Probabilities	Input: Adverse Effects (acute/ late)	Input: HR- QoL	Input: survival	Output: ICER	Output: QALY	Output: LYG	Output: WTP
		final follow-up/ X	single institution data)			oesophagitis, chest pain & pneumonitis					Average LYG for CRT not reporte d	
Marta et al.	Markov model	15 years/ NR	Direct medical costs	X	Constant transition probabilities Time varying transition probabilities using MHD quantifying risk model	Acute & late Xerostomia & dysphagia	X	OS & DFS assumed the same in both arms	31,579 BRL/QALY at 2 years	At 2 years IMRT: 2.03 3DCRT: 1.87 At 15 years IMRT:7.53 3DCRT: 6.37	X	Three times of GDP 72,195BRL/ QALY
Mailhot et al.	Markov model	Lifetime or until 100 years old/1 year	Direct medical costs	3% costs & QALYs	Time varying transition probabilities using quantified MHD Constant 6% risk of cancer death Constant risk for other death	Late effects: Cardiotoxicity Used mean heart dose	X	5-year survival at 94%	NR Results discussed but no figures were reported	QALYs NR but incorporated into the ICER	X	\$50,00/QAL Y \$100,000/ QALY
Li et al.	Markov model	Lifetime or until 80 years old/ 1 year	Direct medical costs	3% costs & QALYs	Constant risk for other death	Late effects: cardiotoxicity Used mean heart dose	X	OS assumed the same in both arms	\$879,729.9/QALY \$/QALY: HRS insertion in hospital: \$96,440 HRS community insertion: \$39,286 Good baseline EF &HRS hospital insertion: \$35,548/QALY Good baseline EF &HRS community insertion: \$9627 Poor baseline EF &HRS hospital insertion: \$341,068 Good baseline EF &HRS community insertion: \$158,439	0.025–0.135 (reported the range)	X	\$37,653/QA LY \$50,00/QAL Y \$100,000/ QALY
Levy et al.	Markov model	5 years/ 3 months	Direct & indirect medical costs	3% costs & QALYs	Varying hazard ratios Equation to calculate transition probabilities	Acute & late grade 2 + GI, GU effects & erectile function	EF in RCT publishe d data	PFS assumed the same in both arms OS based on Kaplan Meir graphs from 2 RCTs	community insertion: \$158,439 ICER: NT\$212,354/ QALY	HRS: 3.599 No HRS: 3.562 PBT resulted in additional 2.61 QALY gains	X	\$100,000/Q ALY Three times the GDP NT \$2157,0 24/QALY

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Table 2 (continued)

Authors [Reference]	CEA model/ Method of EE	Time Horizon/ Cycle Length	Input: Cost data	Input: Discoun ting	Input: Transition Probabilities	Input: Adverse Effects (acute/ late)	Input: HR- QoL	Input: survival	Output: ICER	Output: QALY	Output: LYG	Output: WTP
Lester-Coll et al.	Decision tree & Markov model	2 years/ 1 month	Direct medical costs	3% costs & utilities	Time varying depending on toxicity and survival	Acute & late effects: neurocognitive function & radio- necrosis	X	OS based on Kaplan Meir graphs	ICER 1 brain metastases SRS vs WBRT: \$117,418/QALY SRS vs SRS+ WBRT:\$51,348/ QAL Y SRS+WBRT vs WBRT: \$746,997/QALY ICER SRS at 9 months: \$9917/ QALY at 12 months: \$15,360/ QALY at 24 months: \$33,530/QALY	SRS vs WBRT: 0.94 SRS vs SRS+ WBRT:0.77 SRS+WBRT vs WBRT: 0.79	X	\$100,000/Q ALY (primary) \$300,000/Q ALY (secondary)
Kim et al.	Markov model	5 years/ 1 month	Direct medical costs	3% costs & utilities	Constant transition probabilities Time varying transition probabilities based on trial survival data	Acute & late effects: Neurocognitive function	X	Median OS= 9 months Assumed the same in both arms	QALYs NR but incorporated into the ICER		X	\$100,000/Q ALY (primary) \$50,000 (secondary)
Kerba et al.	Markov model	Lifetime/ 3 months	Direct medical costs	1.5% costs & utilities	Constant transition probabilities	Acute & late effects: Pain, vertebral compression fracture	EORTC QLUC10D	OS based on RCT 12-month survival data	SBRT: \$9040 CAD HRS all: \$63,706AUS/ QALY HSR select: \$20,884AUS/ QALY	SBRT: 1.47 CRT: 1.41	X	\$100,000 CAD/QALY \$28,000 AUS/QALY \$42,000AUS /QALY \$67,000AUS /QALY
Jones et al.	Markov model	5 years/ 6 months	Direct and indirect medical costs	5% costs & utilities	Constant transition probabilities	Acute & late grade 2 + rectal effects	EQ-5D	WHO data for death by any cause	ICER NR- the text only states that partial breast HFRT is the most CE option IMRT vs 3DCRT: \$5164/QALY IMRT vs 2DCRT: \$7072/QALY 3DCRT vs 2DCRT: \$8964/ QALY	HRS all: 8.388 HSR select: 8.381 No HRS: 8.369 QALY gains reported: Whole breast: HFRT vs CFRT 0.05 Partial breast: HFRT vs CFRT 0.07	X	£ 15,00/QAL Y
Glynn et al.	Markov model	50 years/ 1 year	Direct medical costs	3.5% costs & QALYs	Varying transition probabilities (age dependant)	Acute and late skin effects	EQ-5D- 5 L & EQ-5D- 3 L	OS based on multi-centre cohort study OS based on individual patient records and assumed the same in both arms			X	One time the GDP \$1805/ QAL Y
Chauhan et al.	Markov model	5 years/ 3 months for year 1, annually thereafter	Direct medical costs Direct & indirect medical costs using time driven activity- based costing	3% costs & effects	Constant transition probabilities Varying transition probabilities (patient factor dependant)	Acute & late grade 2 + xerostomia	EQ-5D- 5 L		IMRT vs 2DCRT: \$7072/QALY 3DCRT vs 2DCRT: \$8964/ QALY	IMRT: 4.46 3DCRT: 4.39 2DCRT: 4.32	X	
Busschaert et al.	State transition microsimulati on model	20 years/ 1 year	Direct & indirect medical costs	1.5% QALYs 3% QALYs		Late effects Cardiotoxicity Used mean heart dose Acute & Late Grade 2 + Dysphagia & xerostomia Grade	SCORE2 & SCORE2OP	NR	DIBH incremental QALY: 0.04		X	€40,000/ QALY \$250,000/Q ALY (primary)
Brodin et al.	ICER equation	NR/ X	Direct & indirect medical costs	levothy roxine, PEG tube	X		X	OS survival attached to P16 status	IMPT for all patients: \$361,405/QALY	QALYs NR but incorporated into the ICER	X	\$100,000/Q

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Table 2 (continued)

Authors [Reference]	CEA model/ Method of EE	Time Horizon/ Cycle Length	Input: Cost data	Input: Discoun ting	Input: Transition Probabilities	Input: Adverse Effects (acute/late)	Input: HR- QoL	Input: survival	Output: ICER	Output: QALY	Output: LYG	Output: WTP
				depend ency costs		3 + oesophagitis & oral mucositis Hypothyroidism Late effects: Secondary cancer, brainstem necrosis, spinal cord myelitis, tinnitus, blindness, xerostomia, cataracts, endocrine dysfunction						ALY \$500,000/Q ALY
Austin et al.	Markov model	Lifetime/ 1 year	Direct medical costs, indirect non-medical costs (loss of productivity)	3% costs & utilities	Varying transition probabilities based on TCPs, NTCs & DVHs		X	OS based on 5year rates from published study	Mean ICER AUD\$1990/ QALY gained	QALYs calculated for each of the 7 patients and reported as such	X	NR
Austin et al.	Markov model	Lifetime/ 1 year	Direct medical costs	3% costs & QALYs	Varying transition probabilities based on NTCPs, TCPs, SPCIPs and survival data	Late effects: Pneumonitis, cardiotoxicity, secondary cancers	ECOG perform ance status	OS= 55% at 10 years	Average ICER AUD \$84,600	QALYs NR but incorporated into the ICER	X CFRT: 3.25 SBRT: 4.17	Official NICE guidelines £ 20,000 but can negotiated up to £ 30,000
Arruda et al.	Markov model	30 years/ 1 year cycle	Direct medical costs	5% costs & effects	Time varying transition probabilities	X	EQ-5D	OS based on meta-analysis	R\$164.86/ QALY R \$105/LYG	CFRT: 2.00 SBRT: 2.59		R\$25,000/Q ALY

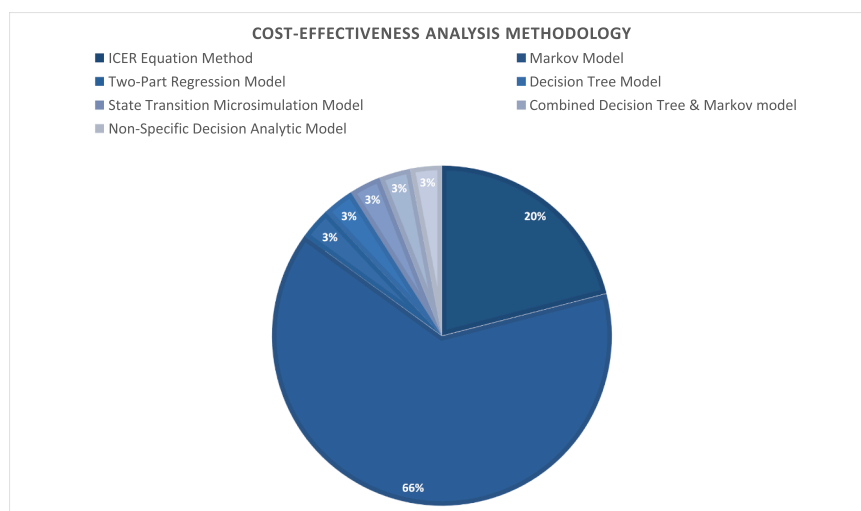


Fig. 2. A pie chart of the distribution of cost-effectiveness analysis methodologies identified in the 35 reviewed studies. Abbreviations: ICER= incremental cost-effectiveness ratio. Note as proportions have been rounded to nearest whole integer, the total is not 100%.

3.3.3. Discounts

In health economics, ‘discounts’ refer to adjusting the value of future costs and health outcomes to present values [94]. Interventions can then be assessed at different time points. Previous literature identified the importance of discounting in health economics as it can influence resource allocation in the health sector [94]. For radiotherapy, discounting costs and health outcomes is essential to accurate and appropriate CEAs. This review identified 3% and 5% as the most common discounts applied, consistent with other health economics research [95]. Six studies (18%) did not apply or report discounts on costs or health outcomes [56,57,59,75,79,80]. Another review found a 70.6% compliance rate in studies applying discount rates to EEs in radiotherapy [23]. In contrast, this review reports a higher compliance rate (82%). However, the review by Nguyen et al. [23] included more studies ($N = 102$) than this review ($N = 35$). Two studies discounted costs only [53,78]. Another study discounted QALYs alone despite applying a 20-year time horizon in the microsimulation model [70]. The extended timeframe warrants discounting costs to adjust future values.

3.3.4. Costing

Costing is one of the two critical components of CEAs. Costs are typically divided into medical and non-medical, further subcategorised as direct or indirect. Direct medical costs include expenses directly related to the intervention (e.g. CT simulation or treatment delivery costs). Indirect medical costs refer to costs associated with managing the condition but are not directly related to the intervention. Late adverse events management is an indirect medical cost of radiotherapy (e.g. long-term beta-blockers for cardiotoxicity). The classification of indirect medical costs lacks clarification, particularly in oncology. Some argue that indirect medical costs should be classified under medical costs only with no sub-categorisation of direct and indirect [96].

Costs of multidisciplinary teams, specialised equipment and facilities should be included in radiotherapy cost assessments [97,98]. Many studies failed to breakdown direct medical costs, limiting financial transparency. For patients, radiotherapy can become financially toxic with several costs involved (e.g. transportation, income loss) [99]. Only three studies included transport costs [53,76,78], and just one accounted for parking fees [53], while most omitted these expenses. There is no agreement in the literature regarding the inclusion of non-medical costs, as some suggest the study perspective as a determinant [100]. In short, this review confirms previous findings that radiotherapy CEAs frequently lack comprehensive cost assessments [16,101], potentially distorting results and influencing funding misallocation.

However, the lack of distinction between payer perspective CEAs and limited societal or societal perspective CEAs in the included literature was striking, the former focused on direct medical costs incurred by insurers or government and the latter on patient time, transport, unpaid caregiver time and the wider productivity loss to society. As radiotherapy techniques advance, comprehensive costing, with separation of these perspectives, is essential to ensure CEAs provide accurate insights for policy decision making.

3.3.5. Health-related Inputs

Adverse events (acute & late) and HR-QoL are key health inputs. Cancer research and treatment no longer focus only on OS and recurrence but rather on the patient, their QoL, and improving their experience. Despite the importance of considering adverse events, the review identified four CEAs (12%) that did not include adverse events [56,73,77,79]. However, twenty two studies included both acute and late events [47, 48, 50, 51, 53–55, 57, 59, 62–69, 75, 76, 80–82], nine reported only late events [49, 58, 60, 61, 70–72, 74, 78], and no study reported only acute events.

Eleven papers limited the grade of adverse events [47,50,53,54,57, 62,67,69,74,80,81]; eight included events >grade 2 [47,50,54,57,62, 67,69,74], two included events >grade 3 [53,80], and one limited some events (dysphagia & xerostomia) to >grade 2 only and other events (oesophagitis & oral mucositis) to >grade 3 events only [81]. Brodin et al. [81] restricted certain events based on the normal tissue complication probability models for oropharyngeal cancer [102]. Patients can experience adverse events across all grades, as reported in RCTs, observational studies and single-institution reports [103–105]. Additionally, management of adverse events is often an out-of-pocket expense for patients, increasing psychosocial and financial toxicities [106]. The associated costs of adverse events and impact on QoL should be reflected in radiotherapy CEAs.

Most studies ($N = 20$) did not include any measure of HR-QoL [47, 49, 50, 53, 54, 57–61, 63–65, 71, 77–82]. Several tools are applied to collate HR-QoL data and incorporate them as utilities into CEAs. The most common tool ($N = 6$) is EQ-5D using data from published literature ($N = 4$) [48,51,69,73], original data ($N = 2$) [68,74] or a combination ($N = 1$) [67]. The EQ-5D tool is general. It does not translate well from one economic landscape to another, as the EQ-5D values can be underestimated [107]. Although the limitations of the EQ-5D are well recognised, it is still reported as the most common tool to extract trial data for EEs [108,109].

The EORTC-QLQ-C30 questionnaire to examine HR-QoL was

mapped onto the EQ-5D algorithm in one study [75], and two studies mapped the EORTC-QLQ-C30 data to the QLU-10D algorithm [56,66]. Other tools used are EPIC with EQ-5D (N = 1) [76], ECOG (N = 1) [72], SCORE-2 + with SCORE-OP (N = 1) [70], and direct reporting of erectile function (N = 1) [62].

3.4. Study outputs

3.4.1. Quantified health outcomes (QALYs & LYG)

Health outcomes are predominantly quantified as quality-adjusted life-years (QALYs) or life-years-gained (LYG). QALYs are the outcome measure recommended by the National Institute for Health and Clinical Excellence (NICE) in the United Kingdom and the National Healthcare Institute in the Netherlands [110,111]. QALYs include both the QoL and quantity of life gained from an intervention. Thus, the inputs such as survival rates (OS or PFS), HR-QoL and adverse events can all be incorporated and measured for CEAs. Although QALYs display advantages supporting their inclusion as an outcome measure for CEAs, they are not without limitations. QALYs fail to encapsulate all aspects of QoL, particularly psychosocial well-being and in palliative settings [112, 113]. Three studies did not use QALYs; two utilised LYG [79,80], and one applied direct xerostomia measures (the primary CEA outcome), excluding both QALYs and LYG [78]. Only one study incorporated both QALYs and LYG [73].

3.4.2. Willingness-to-pay thresholds

The willingness-to-pay (WTP) threshold represents the maximum amount a healthcare system will pay for one additional QALY or LYG. This review noted deviations in the WTP thresholds beyond the two commonly cited in the literature (50,000/QALY, 100,000/QALY) [34]. Fourteen studies (40%) used 50,000/QALY [57,74,76,80], 100,000/QALY [55,62,66,82] or both WTP thresholds [49,50,54,58,60,65]. The remaining studies used other WTP thresholds, such as three times the gross domestic product per the World Health Organisation or country-specific guidelines (e.g. £20,000/QALY) [114]. One study did not report a WTP threshold [47]. The variability across studies suggests that the frequently cited WTP thresholds may not reflect the value of one QALY gained in different health systems. Additionally, variation in WTP thresholds may introduce bias into cost effectiveness results, possibly leading to favouring certain interventions.

3.5. Deterministic sensitivity analysis

The deterministic sensitivity analysis (DSA) is a method of assessing uncertainty in CEAs. Currently, there are no standardised recommendations for which variables to assess in the DSA. This has been argued as an introduction of bias [114]. Despite ongoing debate on the feasibility of DSAs, CEAs need to identify variables that may impact the cost-effectiveness of an intervention. One-way sensitivity analysis was the most frequently used method in 30 (66%) studies [47–51, 53–56, 59–74, 77–79, 81, 82]. This method varies a single parameter at a time and can be conducted on model parameters, costs and outcomes. Although one-way sensitivity analyses are informative, the scope of the analysis is limited as potential interactions between variables are not considered. It is possible to underestimate uncertainties by using one-way sensitivity analysis. A multivariate sensitivity analysis that simultaneously varies two or three variables can bypass this limitation. Combining one-way and two-way sensitivity analysis allows for assessing single variables and the relationships between two variables. Through this methodology, researchers can identify factors that influence CEA outcomes. In radiotherapy, several factors should be considered before implementing a new intervention, including health outcomes, financial investments (construction, staff training, technology) and patient considerations (financial toxicity, informal caregivers, treatment burden). Considering all these factors is often not possible due to incomplete data; thus, performing a multivariate analysis on the

available factors is more methodologically sound than varying a single parameter. Although multivariate analysis has been repeatedly recommended as an improved methodology, only three (9%) studies employed both one-way and two-way sensitivity analysis [57,58,80]. No study simultaneously varied more than two parameters despite its precedence in medical oncology [115,116].

3.6. Probabilistic sensitivity analysis

To conduct a CEA, often several assumptions are part of the methodology due to incomplete or missing information. This can be observed in the transition probabilities, utilities, or health outcomes. A method to account for these uncertainties and mitigate potential bias is to employ a probabilistic sensitivity analysis (PSA). Hatzwell et al. [117] suggested that the 'true' results of CEA models are only identified through PSAs. A Monte Carlo PSA was conducted in 22 studies (%) [48, 49, 55, 57–70, 73, 74, 79, 81, 82], while four (12%) used a non-parametric bootstrap PSA [56,75,76,78]. The most common implementation of Monte Carlo PSAs entails the application of gamma distribution for costs, beta distribution for transition probabilities and a log-normal distribution of continuous parameters, as reported by the 20 studies using Monte Carlo PSAs. The remaining study (N = 1) utilised a triangular distribution for the model parameters due to insufficient empirical data on carbon ion radiotherapy [79]. The authors noted difficulty in applying the other distributions.

The number of simulations in each PSA varied across studies. Of the 21 Monte Carlo PSAs, 10,000 simulations was the most common (N = 10) [48, 62, 63, 66–68, 70, 73, 79, 81]. Others used 999 [69], 1000 [55,57,59], 5000 [58], 50,000 [60,61], and 100,000 [49,64] simulations. Two non-parametric bootstrap PSAs used 1000 simulations [75, 76], one employed 5000 simulations [56], and two did not report the number of simulations [74,78]. The recommended 250,000 simulations for 'true' ICER values were not reached in any study [117], as a maximum of 100,000 simulations (N = 2) was identified [49,64]. This is likely due to practical constraints and the minimal benefit of exceeding 50,000 simulations in CEAs with limited data.

Vanneste et al. [51] did not report the method of PSA or the number of simulations, yet the authors reported a cost-effective result of 77% for a PSA. Additionally, eight studies did not conduct a PSA [47,50,53,54, 71,72,77,80].

4. Discussion

This review aimed to evaluate CEA tools in modern EBRT. First, it is apparent from the synthesis of the available literature that modelling cost effectiveness in EBRT is complex and requires sensitivity to workflow and capacity, adaptability to time points and changing infrastructural needs based on innovation. While included studies included some of these aspects, none captured all. Second, the neglect of the inclusion of health-related quality of life measures, critical to cost effectiveness analysis, indicates the bias that exists towards survival outcomes over patient lived experience in the literature.

This review can report that CEAs in modern EBRT indicate a preference for model-based tools. As previously described, model-based CEAs make several assumptions about disease progression, toxicities and death from disease or other causes. CEA models include assumptions due to poor data availability from the literature. This problem is expected to become more pronounced as radiotherapy is a field with continued innovation, and long-term reliable data for all factors included in CEAs will not always be available. There have been no consistent suggestions within the literature to bypass or resolve the lack of data other than applying assumptions within models. Sampaio et al. [56] used multiple imputation-chained equations to resolve missing HR-QoL data. A limitation of this method is that it operates under the assumption that data are missing at random, which may not be accurate in radiotherapy CEAs.

However, several CEAs were conducted through non-modelling techniques, directly calculating the ICER. Most studies using ICER equations for cost-effectiveness include data from clinical trials; thus, these CEAs have the advantage of complete patient data. Data completeness allows for reliable CEA studies with prolonged follow-up timelines, real-world patient data and often the inclusion of indirect medical and non-medical costs. CEAs based on trial data may reflect future clinical practice and accurately capture cost-effectiveness outcomes (QALYs, LYG, ICERs). Although some argue that real-world cost data is more accurate than trial data [118], its applicability to radiotherapy has not been researched to date. Hence, it is our thesis that direct calculation of the ICER is the most optimal method of CEA in EBRT currently, despite highlighted limitations.

There were several methodological omissions across the selected studies. Brodin et al. [81] discounted QALYs, levothyroxine (long-term thyroid hormone replacement), and PEG tube dependency costs at 3%. The study omitted discounting radiotherapy costs, possibly leading to inaccurate CEA results. In some studies, differentiating methodological omissions from poor reporting practices was difficult. This review recommends adherence to well-established reporting protocols for CEAs (e.g. the Consolidated Health Economics Evaluations Reporting Standards checklist). This conclusion is in agreement with another review of economic evaluations in radiotherapy, which also recommended greater compliance with reporting criteria [23].

The majority of interventions selected for CEAs were for radical (curative) treatment intent, with only five studies including palliative interventions. Four studies focused on brain metastases [55,56,64,65], while only one examined spinal metastases [66]. Approximately over half of radiotherapy patients are treated with palliative intent [119], indicating that a significant portion of resources is allocated to this area of radiotherapy. This imbalance between studies on radical radiotherapy and palliative radiotherapy is reflective of the underrepresentation of palliation across the radiotherapy literature, previously acknowledged in a bibliometric analysis of global radiotherapy research, where palliative radiotherapy accounted for only 5% of research outputs [120]. In this review, the identified studies did not thoroughly examine palliative radiotherapy techniques or assess the

cost-effectiveness of palliative radiotherapy. CEAs are particularly advantageous as they incorporate both economic and patient-centred data, making the evaluation of palliative radiotherapy’s cost-effectiveness not only feasible but also relevant.

Although this review recommends standardising CEA tools in radiotherapy, it acknowledges implementation challenges. Variability of discount rates and WTP thresholds highlight this issue, as some studies follow country-specific guidelines (e.g. NICE), reflecting jurisdictional value pluralism. Standardisation would, of course, enhance international comparability, particularly for rare malignancies or new treatment techniques but would likely be difficult as full transparent reporting would be required.

The review concentrated solely on radiotherapy interventions. One limitation of this review is that radiotherapy is often prescribed as part of a multimodality treatment. This limitation highlights the necessity for future research on CEA methodologies of combined treatments (e.g. chemoradiation).

5. Conclusion

The breadth of methods and adaptations in conducting CEAs suggests that existing practices do not fully address radiotherapy needs. Despite emphasis on the importance of CEAs in radiotherapy, the evidence base offers little consensus. This review underlines areas of contention, dispute and agreement in EBRT CEAs and lays the groundwork for further research in this area.

CRedit authorship contribution statement

Taimia Alkanzi: Writing – original draft, Investigation, Formal analysis. Michelle Leech: Writing – review & editing, Supervision, Conceptualization.

Declaration of Competing Interest

The authors confirm that they have no conflicts of interest to declare.

APPENDICES.

Literature search strings for initial search of 3 databases, PubMed, Embase & Scopus. Each database was immediately limited to papers published in 2014–2024

Database	Search Strings
PubMed	((‘Cost effectiveness’) AND (‘radiotherapy’ OR ‘radiation therapy’ OR ‘radiation oncology’))
	((‘Cost effectiveness analysis’) AND (‘radiotherapy’ OR ‘radiation therapy’ OR ‘radiation oncology’))
	((‘Cost utility analysis’) AND (‘radiotherapy’ OR ‘radiation therapy’ OR ‘radiation oncology’))
	((‘Markov model’) AND (‘radiotherapy’ OR ‘radiation therapy’ OR ‘radiation oncology’))
	((‘Monte Carlo model’) AND (‘radiotherapy’ OR ‘radiation therapy’ OR ‘radiation oncology’))
	((‘Economic evaluation’) AND (‘radiotherapy’ OR ‘radiation therapy’ OR ‘radiation oncology’))
	((‘Health economics’) AND (‘radiotherapy’ OR ‘radiation therapy’ OR ‘radiation oncology’) AND (‘ICER’ OR ‘QALY’ OR ‘LYG’ OR ‘WTP’))
	((‘Cost effectiveness’) AND (‘radiotherapy’ OR ‘radiation therapy’ OR ‘radiation oncology’))
	((‘Cost effectiveness analysis’) AND (‘radiotherapy’ OR ‘radiation therapy’ OR ‘radiation oncology’))
	((‘Cost utility analysis’) AND (‘radiotherapy’ OR ‘radiation therapy’ OR ‘radiation oncology’))
Embase	((‘Markov model’) AND (‘radiotherapy’ OR ‘radiation therapy’ OR ‘radiation oncology’))
	((‘Monte Carlo model’) AND (‘radiotherapy’ OR ‘radiation therapy’ OR ‘radiation oncology’))
	((‘Economic evaluation’) AND (‘radiotherapy’ OR ‘radiation therapy’ OR ‘radiation oncology’))
	((‘Health economics’) AND (‘radiotherapy’ OR ‘radiation therapy’ OR ‘radiation oncology’) AND (‘ICER’ OR ‘QALY’ OR ‘LYG’ OR ‘WTP’))
	((‘Cost effectiveness’) AND (‘radiotherapy’ OR ‘radiation therapy’ OR ‘radiation oncology’))
	((‘Cost effectiveness analysis’) AND (‘radiotherapy’ OR ‘radiation therapy’ OR ‘radiation oncology’))
	((‘Cost utility analysis’) AND (‘radiotherapy’ OR ‘radiation therapy’ OR ‘radiation oncology’))
	((‘Markov model’) AND (‘radiotherapy’ OR ‘radiation therapy’ OR ‘radiation oncology’))
	((‘Monte Carlo model’) AND (‘radiotherapy’ OR ‘radiation therapy’ OR ‘radiation oncology’))
	((‘Economic evaluation’) AND (‘radiotherapy’ OR ‘radiation therapy’ OR ‘radiation oncology’))
Scopus	((‘Health economics’) AND (‘radiotherapy’ OR ‘radiation therapy’ OR ‘radiation oncology’) AND (‘ICER’ OR ‘QALY’ OR ‘LYG’ OR ‘WTP’))

Appendix B Table 1

Drummond Checklist for 35 identified cost-effectiveness analyses studies. The Drummond 10-point checklist allows for a critical appraisal of economic evaluations

Authors [Ref]	Research Question well defined?	Comprehensive description of alternatives?	Effectiveness of program established?	Important & relevant costs & consequences for each alternative identified?	Costs & consequences measured accurately & appropriately?	Costs & consequences valued credibly?	Costs & consequences adjusted for differential timing?	Incremental analysis of costs & consequences preformed?	Allowance made for uncertainty estimates?	Presentation & discussion of study results include all issues of concern to users?	Score
Zhou et al.	✓	✓	✓	X	✓	✓	✓	✓	✓	✓	9/10 good
Yong et al.	✓	✓	✓	X	✓	✓	✓	✓	X	X	7/10 average
Yang et al.	✓	✓	✓	X	✓	✓	✓	✓	✓	✓	9/10 good
Xie et al.	✓	✓	✓	X	✓	X	✓	✓	✓	✓	8/10 good
Vanni et al.	✓	✓	✓	X	✓	X	✓	✓	X	✓	7/10 average
Vanneste et al.	✓	✓	✓	X	✓	✓	✓	✓	✓	✓	9/10 good
Sun et al.	✓	✓	✓	X	✓	X	X	✓	✓	✓	7/10 average
Sprave et al.	✓	✓	✓	X	✓	X	✓	✓	X	✓	9/10 average
Sher et al.	✓	✓	✓	X	✓	✓	✓	✓	✓	✓	7/10 good
Shah et al.	✓	✓	✓	X	✓	✓	X	✓	X	✓	8/10 average
Schumacher et al.	✓	✓	✓	X	✓	✓	✓	✓	X	✓	8/10 good
Savitz et al.	✓	✓	✓	X	X	✓	✓	✓	✓	✓	8/10 good
Sampaio et al.	✓	✓	✓	X	✓	✓	X	✓	✓	✓	8/10 good
Richard et al.	✓	✓	✓	X	✓	✓	X	✓	✓	✓	9/10 good
Qu et al.	✓	✓	✓	X	✓	✓	✓	✓	✓	✓	9/10 good
Parthan et al.	✓	✓	✓	X	✓	✓	✓	✓	✓	✓	7/10 good
Perrier et al.	✓	✓	✓	X	✓	X	X	✓	✓	✓	8/10 average
Okazaki et al.	✓	✓	✓	X	✓	✓	X	✓	✓	✓	7/10 good
Mitera et al.	✓	✓	✓	X	✓	✓	X	✓	X	✓	6/10 average
Marta et al.	✓	✓	✓	X	✓	X	X	✓	✓	X	8/10 average
Mailhot et al.	✓	✓	X	X	✓	✓	✓	✓	✓	✓	6/10 good
Li et al.	✓	✓	X	X	X	X	✓	✓	✓	✓	8/10 average
Levy et al.	✓	✓	✓	X	✓	X	✓	✓	✓	✓	8/10 good
Leung et al.	✓	✓	✓	X	✓	X	✓	✓	✓	✓	9/10 good
Lester-Coll et al.	✓	✓	✓	X	✓	✓	✓	✓	✓	✓	9/10 good
Kim et al.	✓	✓	✓	X	✓	✓	✓	✓	✓	✓	7/10 good
Kerba et al.	✓	✓	✓	X	X	X	✓	✓	✓	✓	8/10 average
Jones et al.	✓	✓	✓	X	✓	X	✓	✓	✓	✓	7/10 good
Glynn et al.	✓	✓	✓	X	X	X	✓	✓	✓	✓	8/10 average
Chauhan et al.	✓	✓	✓	X	✓	✓	✓	✓	✓	✓	8/10 good
Busschaert et al.	✓	✓	✓	X	✓	✓	X		✓	✓	8/10 good
Brodin et al.	✓	✓		X	✓	✓	X		✓	✓	6/10 good
✓✓Austin et al.	✓	✓	✓	X	✓	X	✓	✓	X	✓	6/10 average
Austin et al.	✓	✓	X	X	✓	X	✓	✓	X	✓	6/10 average

(continued on next page)

Appendix B Table 1 (continued)

Authors [Ref]	Research Question well defined?	Comprehensive description of alternatives?	Effectiveness of program established?	Important & relevant costs & consequences for each alternative identified?	Costs & consequences measured accurately & appropriately?	Costs & consequences valued credibly?	Costs & consequences adjusted for differential timing?	Incremental analysis of costs & consequences preformed?	Allowance made for uncertainty estimates?	Presentation & discussion of study results include all issues of concern to users?	Score
Arruda et al.	✓	✓	✓	X	✓	X	✓	✓	✓	✓	8/10 good 271/
Final scores	35/ 35 = 100%	35/35 = 100%	32/35 = 91%	0/35 = 0%	31/35 = 89%	20/35 = 57%	25/35 = 71%	35/35 = 100%	27/ 35 = 77%	31/35 = 89%	350 = 77%

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